

Experiment Results

Test 7 fourth-order follow-up

Run run-29c4c4228ffddb4; analysis analysis-0001-ac36b679; classification pass.
Unchanged SS OCF 1 action and Test 6 clock calibration. Original Test 7 remains failed.
Prior Test 7 comparison is an inspected-data diagnostic; held-out wave is the prospective check.
No two-object recoil, coordinate invariance, gravity, angular stability, or emergent spacetime is tested.

Held-out interval: actual $-2.087776737e-07$; order-two $-2.093903858e-07$; order-four $-2.089527184e-07$ cycles.

Numerical interval budget $1.295e-09$ cycles.

conservation: pass
heldout-interval: pass
heldout-waveform: pass
marker-timing: pass
prediction-first: pass
record-resolution: pass
sign-even: pass

prior-diagnostic

Question: Does the derived fourth-order coefficient explain the inspected residual sign and scaling?

Reading: Compare the saved Test 7 amplitude ladder at the same nominal $A=.004$ marker times.

Significance: This diagnoses the missing order independently of a fitted coefficient.

Limitation: The prior histories and marker times were already inspected; this is not a prospective prediction.

heldout-phase

Question: Does the locked fourth-order prediction improve a new local clock history?

Reading: Held-out finest trace error: second 0.1882% , fourth 0.1712% .

Significance: A separate waveform tests the coefficient after the inspected-history diagnostic.

Limitation: Quiet phase at the pulse markers is counterfactual; grid differences and inversion error remain.

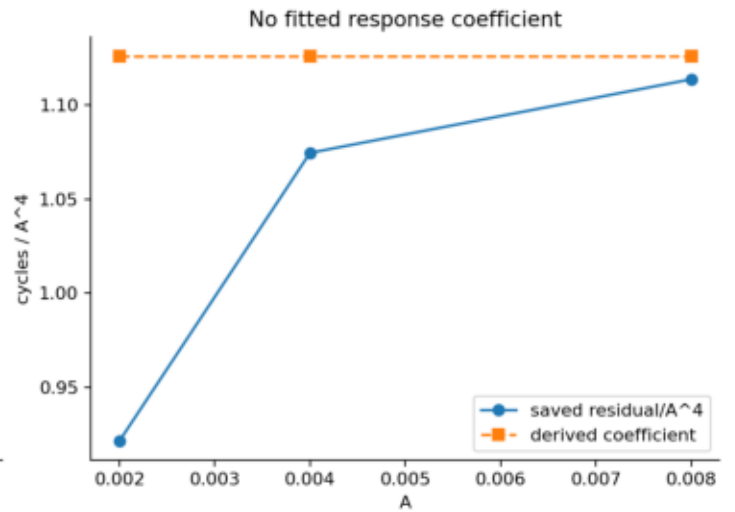
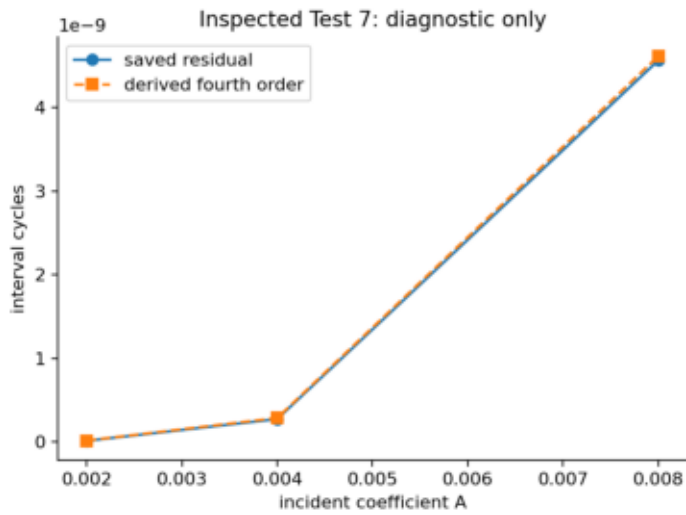
marker-reconstruction

Question: Do derivative-informed markers and twice-frequent output explain local timing error?

Reading: On $h=.025$, compare markers at 0.05 and decimated 0.1 ; coarse vacuum/known-Z maximum errors are $0.05998/0.05981$.

Significance: Separates temporal interpolation from surface-propagation error.

Limitation: The coarse vacuum control is post-hoc; Hermite cannot repair spatial dispersion or WKB error.

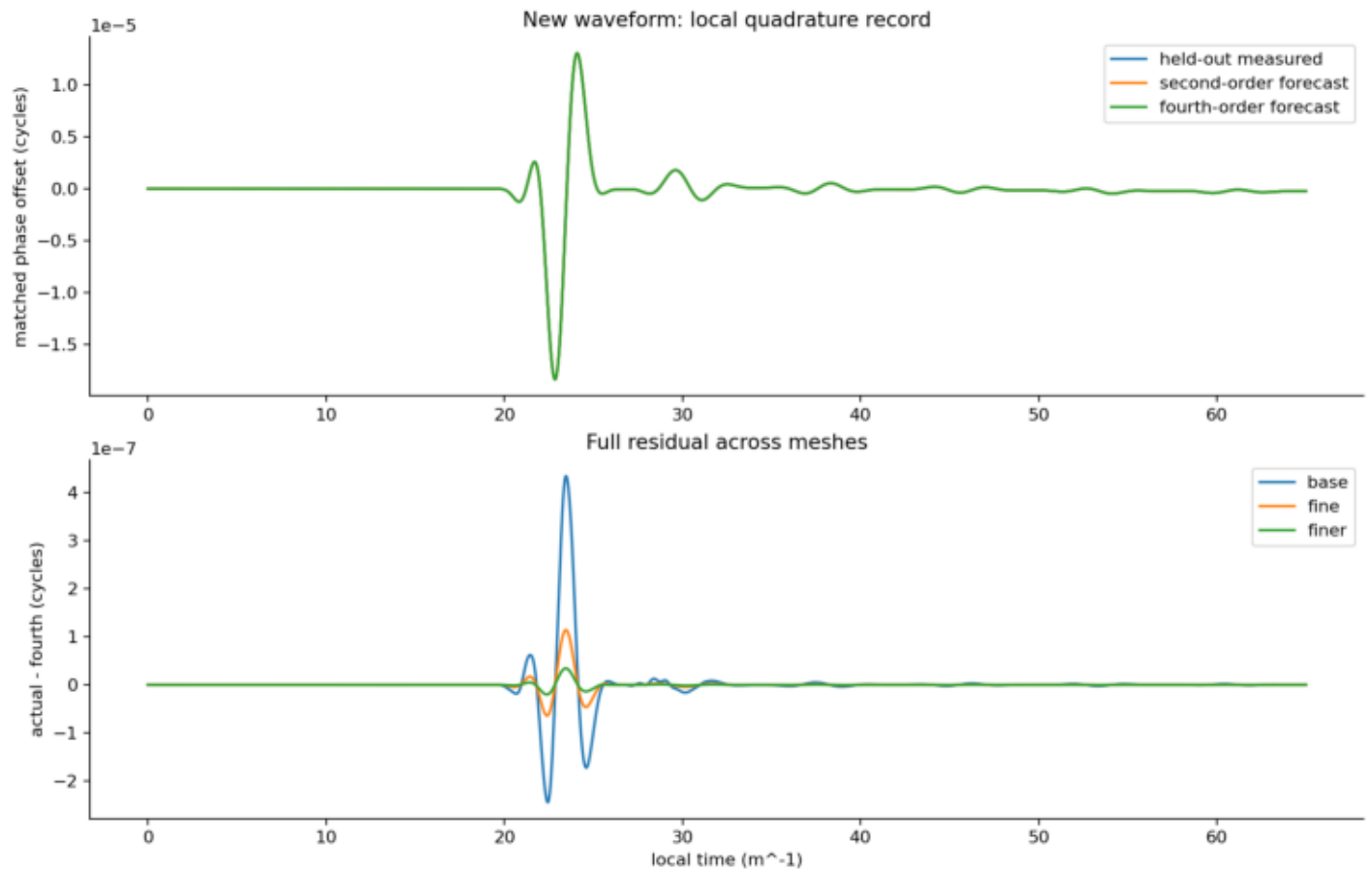


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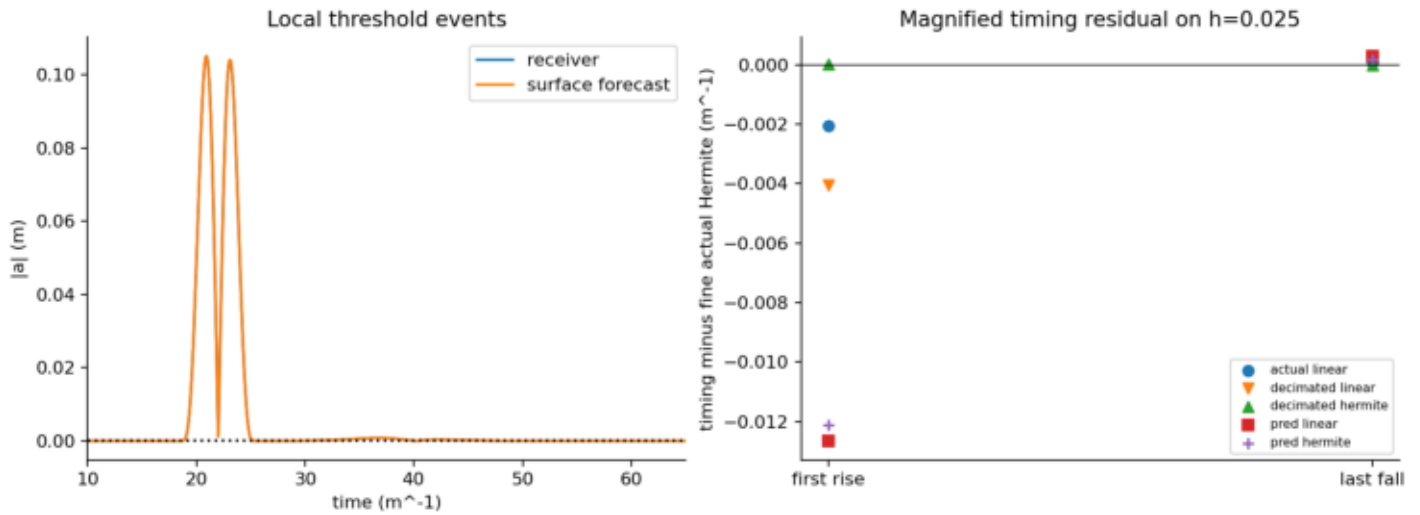


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